

Python Data Types Theory Questions Quiz

1. What is the difference between a list and a tuple in Python?

A) Lists are mutable, tuples are immutable

Correct Answer: A

Explanation: The main difference is that lists can be modified after creation (mutable) while tuples cannot (immutable).

2. Explain the concept of mutability with respect to Python data types.

A) It determines whether an object can be changed after creation

Correct Answer: A

Explanation: Mutability refers to whether an object's contents can be altered after it's created.

3. What happens when you assign one list to another variable and then modify the original list?

B) Both variables reference the same list, so both show changes

Correct Answer: B

Explanation: In Python, assignment creates a new reference to the same object. This is called aliasing.

4. How is a tuple different from a list in terms of usage and performance?

B) Tuples are faster and used for fixed data

Correct Answer: B

Explanation: Tuples are generally faster than lists because of their immutability.

5. Why would you use a tuple instead of a list?

A) When you need an immutable sequence

Correct Answer: A

Explanation: Tuples are used when you want to ensure data can't be modified or for hashable keys.

6. Describe how dictionaries store data in key-value pairs.

B) Using a hash table implementation

Correct Answer: B

Explanation: Dictionaries use hash tables for fast lookups.

7. Can a dictionary have duplicate keys?

B) No, each key must be unique

Correct Answer: B

Explanation: Dictionary keys must be unique to avoid ambiguity.

8. How can you update the value of a key in a dictionary?

B) By assigning a new value to the key

Correct Answer: B

Explanation: Use `my_dict[key] = new_value` to update a value.

9. What will happen if you try to access a key that doesn't exist in a dictionary?

C) It raises a KeyError

Correct Answer: C

Explanation: Direct access to a non-existent key raises a KeyError.

10. What are some common methods used with dictionaries?

B) `keys()`, `values()`, `items()`, `get()`

Correct Answer: B

Explanation: These methods allow accessing and safely retrieving data.

11. What is a set in Python and how is it different from a list?

B) A set is an unordered collection of unique elements

Correct Answer: B

Explanation: Sets are unordered and automatically remove duplicates.

12. How does Python handle duplicate values in a set?

B) It automatically removes duplicates

Correct Answer: B

Explanation: Duplicates are removed automatically in sets.

13. What methods can be used to add or remove items in a set?

B) `add()` and `remove()`

Correct Answer: B

Explanation: Sets use `add()` and `remove()`.

14. Can you index or slice a set?

B) No, because sets are unordered

Correct Answer: B

Explanation: Sets do not support indexing or slicing.

15. Explain the difference between `.remove()` and `.discard()` in sets.

B) `.remove()` raises `KeyError` if element doesn't exist, `.discard()` doesn't

Correct Answer: B

Explanation: `discard()` won't raise an error if the item doesn't exist.

16. What are some common use cases for using sets in Python?

B) For membership testing and eliminating duplicates

Correct Answer: B

Explanation: Sets are great for membership tests and deduplication.

17. How do you loop through a dictionary and access both keys and values?

B) Using the `.items()` method

Correct Answer: B

Explanation: `.items()` returns key-value pairs.

18. How can you convert a list into a set and why would you want to do that?

B) `set(list)`, to remove duplicates

Correct Answer: B

Explanation: Useful for removing duplicates.

19. Is it possible to store a list inside a tuple? Explain.

B) Yes, but the list remains mutable

Correct Answer: B

Explanation: Tuples can hold mutable types, and lists inside them remain mutable.

20. Describe the impact of mutability on memory usage and performance in Python.

B) Immutable objects can be more memory efficient for some operations

Correct Answer: B

Explanation: Python can optimize memory for immutable types.